

# Question Paper

Exam Date & Time: 16-Jun-2025 (09:30 AM - 12:30 PM)



## MANIPAL ACADEMY OF HIGHER EDUCATION

SECOND SEMESTER B.TECH. DEGREE EXAMINATIONS - JUNE 2025  
SUBJECT: CSE 1071/CSE\_1071 - PROBLEM SOLVING USING COMPUTERS  
(MAKE-UP)

Marks: 50

Duration: 180 mins.

Answer all the questions.

1A) Explain in brief the memory unit of a computer system. (3)

Course Outcome: "CO1" / Cognitive Level: "Understand"

1B) Discuss the Type Cast operator in C language with an example. Evaluate the following expression by clearly showing all the intermediate steps. (3)  
 $-10 * (2 - 3) / 15 \% (5 + 7) + 8 * 6 / 12$

Course Outcome: "CO2" / Cognitive Level: "Application"

1C) Write a program in C to print a number in words using switch case statement. (4)  
Example:  
Enter the number: 5327  
Five Three Two Seven

Course Outcome: "CO2" / Cognitive Level: "Application"

2A) Compare between while and do-while statements in C using appropriate code snippets. (2)

Course Outcome: "CO2" / Cognitive Level: "Understand"

2B) i) With the help of a C program, show the limitation of the traditional **scanf( )** function in reading a string variable in C programming. Show the program's sample input/output. (4)  
ii) Using the same C program, show how this limitation can be overcome by using a different function. Show the program's sample input/output.

Course Outcome: "CO3" / Cognitive Level: "Application"

2C) Develop and write a C program to count the number of positive, negative and zero elements in a 2D array. The program should prompt the user to enter the elements of the array. Finally, it should display the count of positive, negative, and zero elements and the sum of all the elements in the matrix. (4)

Course Outcome: "CO3" / Cognitive Level: "Application"

3A) Write a function prototype for a function named **calculate** that takes three parameters: an integer, a floating-point number, and a character. The function should perform different operations based on the character ('a' for addition, 's' for subtraction, 'm' for multiplication) and return the result as a floating-point number. Then, define the function **calculate** to perform the operations as mentioned (2)

above. Note: Main function not required.

Course Outcome: "CO4" / Cognitive Level: "Application"

- 3B) Define a function named **findAverage()** that takes an array of integers and its size as parameters and returns the average of the array elements. Write another function named **displayAverage()** that takes an array of integers and its size, calls **findAverage()**, and prints the average. Discuss the scope of variables in both functions with the complete program. (4)

Course Outcome: "CO4" / Cognitive Level: "Application"

- 3C) Write a recursive function in C to **count the number of zeroes** in a non-negative integer. The **main()** function must take the user input, call the recursive function and print the result in **main()**. (4)  
Sample I/O:  
Enter a non-negative integer: 10340020  
Number of zeroes = 4

Course Outcome: "CO4" / Cognitive Level: "Application"

- 4A) A student wants to write a program that increments two variables by 1. To modify the original values in the calling function, they decide to pass the variables by reference using pointers. (3)  
i) Write a **C function** **void increment(int \*a, int \*b)** that increments the values of **a** and **b** by 1.  
ii) In the **main()** function, declare two integers **x = 5** and **y = 10**, and call **increment()** to modify their values. Print the updated values of **x** and **y**.

Course Outcome: "CO4" / Cognitive Level: "Application"

- 4B) Write a C program to demonstrate the nesting of functions to find the sum and product of integers as defined below: (4)  
Function **sum** should take two integers and return their sum.  
Function **multiply** should take three integers, call the function **sum** to compute the sum of the first two integers, and return the product of the sum and the third integer.  
In main, read three integers, call the function **multiply** and display the result.

Course Outcome: "CO4" / Cognitive Level: "Application"

- 4C) A library management system needs to store information about books including the title, author name, and price. (3)  
Define a structure **Book** with members **title**, **author**, and **price**. Write a C program to input and display details of a single book using the structure. (Add some computation)

Course Outcome: "CO5" / Cognitive Level: "Application"

- 5A) An organization wants to maintain employee details including name, ID, and address. The address itself has components like city, state, and pin code. This calls for a structure inside a structure. (3)  
Define a structure **Address** with members **city**, **state**, and **pin**, and a structure **Employee** with members **name**, **id**, and **address** (of structure type **Address**).  
Write a C program to input and display the data for one employee using the above structures. Write a function **void displayEmployee(struct Employee e)** to print the employee details.

Course Outcome: "CO5" / Cognitive Level: "Understand"

- 5B) i) Explain how pointers can be used to access array elements in C. (4)  
ii) Write a C program that takes number of elements and an array of integers as input and prints the elements using both array notation and pointer notation.  
iii) Using pointer notation only, **print the array in reverse order**.

Course Outcome: "CO5" / Cognitive Level: "Application"

- 5C) Explain the common signs of a **phishing attack** that users should look out for. Discuss the (3)

measures that organizations can implement to **train employees** and reduce the risk of falling for phishing scams?

Course Outcome: "CO6" / Cognitive Level: "Application"

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| Event Name                                                                        |                | Subject                                      |       |
|-----------------------------------------------------------------------------------|----------------|----------------------------------------------|-------|
| MIT MPL and BLR- BTech-MSc-MCA - II-IV and VI Sem - Makeup Examination - Jun 2025 |                | CSE 1071-CHM:PROBLEM SOLVING USING COMPUTERS |       |
| #                                                                                 | Course Outcome | Description                                  | Marks |
| 1                                                                                 | 1              | CO 1                                         | 3     |
| 2                                                                                 | 2              | CO 2                                         | 9     |
| 3                                                                                 | 3              | CO 3                                         | 8     |
| 4                                                                                 | 4              | CO 4                                         | 17    |
| 5                                                                                 | 5              | CO 5                                         | 10    |
| 6                                                                                 | 6              | CO 6                                         | 3     |
| Total                                                                             |                |                                              | 50    |

\* **NA** - CO Not Selected

| Event Name                                                                        |             | Subject                                      |           |
|-----------------------------------------------------------------------------------|-------------|----------------------------------------------|-----------|
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| #                                                                                 | Level       | No of Questions                              | Marks (%) |
| 1                                                                                 | Remember    | 0                                            | 0(0%)     |
| 2                                                                                 | Understand  | 3                                            | 8(16%)    |
| 3                                                                                 | Application | 12                                           | 42(84%)   |
| 4                                                                                 | Analysis    | 0                                            | 0(0%)     |
| 5                                                                                 | Create      | 0                                            | 0(0%)     |
| 6                                                                                 | Evaluate    | 0                                            | 0(0%)     |
| Total                                                                             |             | 15                                           | 50 (100%) |